

Layer

KV memory :

(key, value)

KV_{memo}

$$= \frac{2 \cdot \text{Batch} \cdot \text{seq len} \cdot \text{Layers} \cdot kv \text{ heads} \cdot \text{head dim} \cdot \text{bytes}}{T \text{ Blocks}}$$

Format
FP8
int8
etc

* Since (seq-len) $\Rightarrow$ multiplier $\rightarrow$ Doubling the context $\rightarrow$ drastically increases the VRAM to the

Long context is hard

128k context window means it can accept 128k tokens

- System degradation: higher cost, mem pressure, slower schedul

- Quality collapse: model attention become diluted
"lost in the middle" (forgetting facts in the middle)

- Security, accuracy risks

IT DOESN'T MEAN IT WILL BE RELIABLE

PREFIX
caching

- allow to reuse existing KV cache for prompts prefixes
- are shared

- Repeated, static prompts
- Multi turn chat sessions
- Few Shot examples

RAG vs Long Context

Long
context

• when every piece of context is relevant

↳ order matters, context change per user request
• retriever consistently misses critical details

Rag

• massive corpus of data, only slice of the data matter to the user query, up-to-date
freshness, strict citations, enforce access doc permission